approach is called *copy-on-write*: a copy of the original item is created in the upper layer, and changes are applied to the copy rather than to the original item.

When an image is *run* as a container, one last layer is automatically applied on top of all the existing ones: the *container layer*. It is the only writable layer, and it's used to store data created during the execution of the container itself. At runtime, this layer might be used to generate files required by the application to run or maybe to store temporary data. Even though it's writable, remember that it's volatile: once you delete your container, everything stored in that layer is gone. Figure 6.3 compares the layers in a running container and those in the corresponding image.

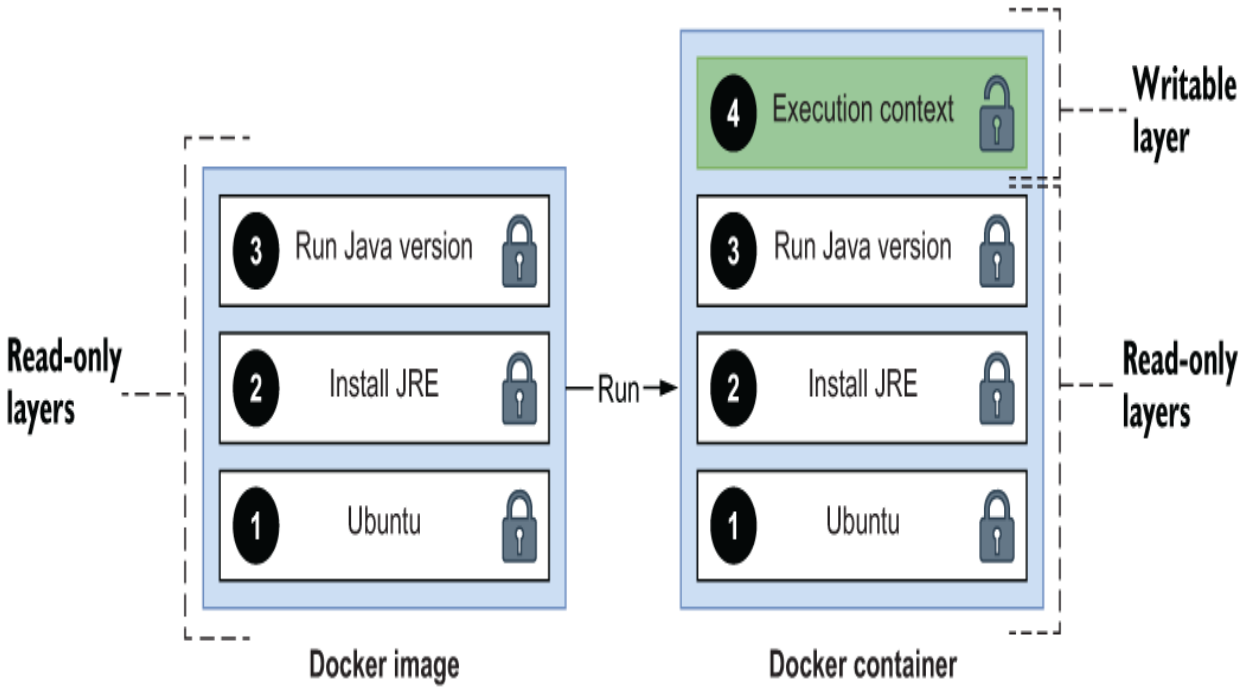


Figure 6.3 Running containers have an extra layer on top of the image layers. That is the only writable layer, but remember that it's volatile.

NOTE The fact that all the layers in a container image are read-only has some security implications. You should never store secrets or sensitive information in the lower layers because they will always be